

Chapter 11 — Building the *Kriyā* (क्रिया): Sanskrit's Molecule

11.1 From Semantic Atom to Verbal Molecule

Chapter 10 established the धातु: (*dhātuh*) as Sanskrit's semantic atom. The atom contains core meaning, but Sanskrit does not use a *dhātuh* directly inside a sentence. [NOTE: apadam-constraint] Like an atom in chemistry, it must combine before it can function as a word. Additional sonomers attach to it, and it becomes a verb, a noun, or an adjective. This chapter follows one of those routes: the engineering through which a *dhātuh* becomes a completed verb.

For example, «इ» (*i*) is a *dhātuh* meaning *to go*. It does not yet tell the listener who goes or how many people go. It also lacks the ending required to function as a verb in a sentence. एति (*eti*) is one verb built from that atom. It means *he, she, or it goes*.

Sanskrit builds this information into the completed verb as part of the engineering that permits free word order. The verb identifies whether the action belongs to the speaker, the person addressed, or someone being described. It also identifies how many perform the action. Other words receive endings that identify their roles in the sentence. The sentence therefore carries its grammatical relations inside the completed words instead of assigning them through a fixed sequence.

The Vedas use this freedom extensively. Because endings identify each word's role, a mantra can place its words where they serve meter, resonance, emphasis, and memory. Once the mantra was seen and entered transmission, its received order became invariant. Chapter 12 demonstrates this freedom at the scale of a complete sentence.

Sanskrit forms a completed verb by preparing the atom and adding a verbal ending. That preparation may leave the atom almost unchanged. It may change a vowel, insert another sound, repeat part of the atom, or add material at a restricted position. The ending then tells the listener who performs the action and how many perform it. Other formations add time, command, possibility, desire, and further distinctions.

These changes create an engineering problem. Sanskrit must change the atom enough to make it usable without destroying its identity. A learner should be able to begin with the completed verb, separate the ending and the material used during preparation, and identify the *dhātuh* from which the verb was formed. The completed verb must therefore preserve enough of its construction for the learner to trace it back to its atom.

The Vedas preserve completed verbs built through these procedures. We can therefore examine the verbs first, identify what changed inside each one, and only then turn to the grammatical analysis that Pāṇini made explicit. [NOTE: vedic-kriyapadas-

before-panini]

11.2 Five Vedic Activation Procedures

The first five examples come from the Ṛgveda. Each one shows a different way in which Sanskrit prepares a *dhātuh* and completes it as a क्रियापदम् (*kriyāpadam*), a verb ready for sentence use.[NOTE: rigvedic-kriya-examples]

Each procedure remains available in *laukika* Sanskrit. Its presence in a received Vedic passage therefore gives later speakers and students an invariant example against which they can check both formation and pronunciation.

This chapter calls that preparation **activation**. The atom already contains meaning. Activation changes or extends the atom in the specific way required before a verbal ending can join it. The completed verb can then express an action inside a sentence.

Vowel Change: ⟨इ⟩ (*i*) → एति (*eti*)

चन्द्रमाः । नक्तम् । एति ।

candramāḥ | *nakṭam* | *eti* |

RV 1.24.10b

The atom is ⟨इ⟩ (*i*), a one-*mātrā* vowel atom meaning *to go*. Sanskrit changes इ (*i*) into ए (*e*) before the ending ति (*ti*) completes the verb.

⟨इ⟩ (*i*) → ए (*e*) + ति (*ti*) → एति (*eti*)

The completed एति (*eti*) means *he, she, or it goes*. The atom remains identifiable even though its vowel has changed. A learner can account for the change and retrace the path from the completed verb to its atom. That recoverable path is more important than a letter-for-letter resemblance between the two forms.

Vowel Change and Insertion: ⟨भू⟩ (*bhū*) → भवति (*bhavati*)

क्रतुः । भवति । उक्थ्यः ।

kratuh | *bhavati* | *ukthyah* |

RV 1.17.5c

The atom is ⟨भू⟩ (*bhū*), meaning *to become* or *to come into being*. Sanskrit prepares this atom through more than one step. The long ऊ (*ū*) first changes to ओ (*o*). Because another vowel follows, ओ then takes the pronounceable form अव् (*av*). Sanskrit places अ (*a*) after this prepared form, and ति completes the verb.

⟨भू⟩ (*bhū*) → भो (*bho*) → भव् (*bhav*) + अ (*a*) + ति (*ti*) → भवति (*bhavati*)

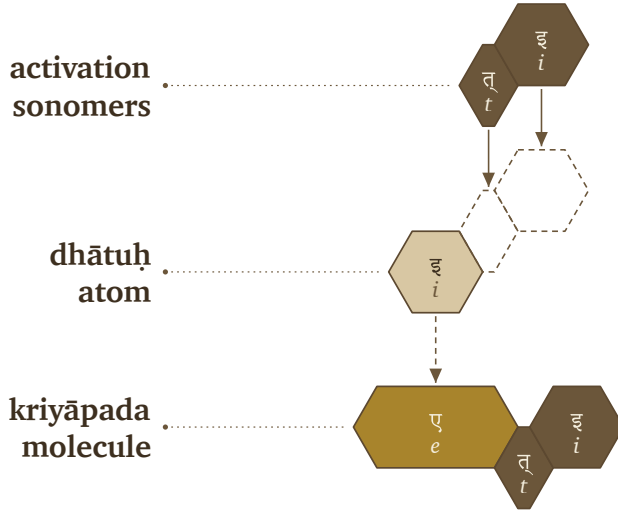


Figure 11.1 — Vedic assembly: «इ» (*i*) becomes एति (*eti*).

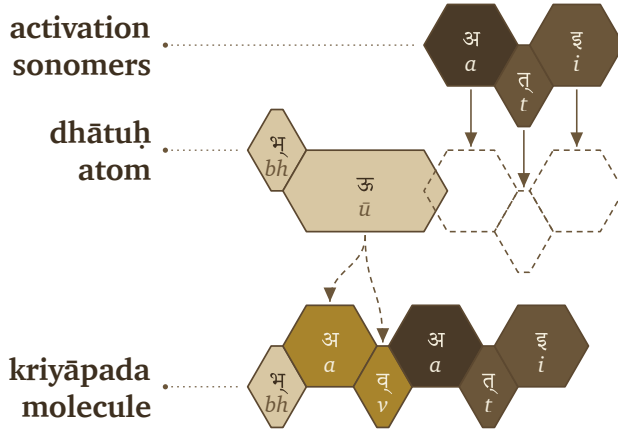


Figure 11.2 — Vedic assembly: «भू» (*bhū*) becomes भवति (*bhavati*).

The completed भवति (*bhavati*) means *he, she, or it becomes*. Its surface form differs considerably from «भू», but every difference follows a restricted sequence. A learner can reverse the sequence and return from भवति to «भू».

Repetition: «दा» (*dā*) → ददाति (*dadāti*)

सोमः । धेनुम् । सोमः । वीरम् । ददाति ।

somaḥ | *dbenum* | *somaḥ* | *vīram* | *dadāti* |

RV 1.91.20a

The atom is «दा» (*dā*), meaning *to give*. The completed verb does not attach ति directly to the bare atom. Sanskrit first repeats the opening consonant with a short अ. The original long vowel remains in the next part of the form.

«दा» (*dā*) → द (*da*) + दा (*dā*) + ति (*ti*) → ददाति (*dadāti*)

The completed ददाति (*dadāti*) means *he, she, or it gives*. The repetition is compact, but it is not decorative. It belongs to the formation procedure. The listener can hear both the repeated opening and the «दा» atom within the completed verb.

Nasal Insertion: «रुध्» (*rudh*) → रुणद्धि (*ruṇaddhi*)

देवकामः । न । रुणद्धि ।

devakāmaḥ | *na* | *ruṇaddhi* |

RV 10.42.9c

The atom is «रुध्» (*rudh*), meaning *to obstruct, restrain, or block*. Here Sanskrit places a nasal sound inside the atom, before its final consonant. The sounds at the new boundary then adjust as the ending joins.

«रुध्» (*rudh*) → रुणध् (*ruṇadh*) + ति (*ti*) → रुणद्धि (*ruṇaddhi*)

The completed रुणद्धि (*ruṇaddhi*) means *he, she, or it obstructs or restrains*. The nasal sound enters the middle of the atom. The completed form still retains the opening रु and the final ध्, so the inserted sound changes the atom without concealing it.

Nasal Extension: «सु» (*su*) → सुनोति (*sunoti*)

सोमम् । सुनोति । भवति । द्युमान् । अह ।

somam | *sunoti* | *bhavati* | *dyumān* | *aha* |

RV 5.34.3a

The atom is «सु» (*su*), meaning *to press or express Soma*. This procedure adds nasal-bearing material after the atom rather than placing a nasal sound inside it. In this completed form, the extension appears as नो (*no*) before ति.

«सु» (*su*) + नो (*no*) + ति (*ti*) → सुनोति (*sunoti*)

The completed सुनोति (*sunoti*) means *he, she, or it presses Soma*. Compare it with रुणद्धि. Both verbs contain an added nasal sound, but Sanskrit places that sound differently. रुणद्धि inserts it inside the atom. सुनोति extends the atom after its final sound.

Figure 11.3 places the five procedures beside one another.

ATOM	PREPARATION	COMPLETED VERB
《इ》 i	vowel change	→ एति eti
《भू》 bhū	vowel change and insertion	→ भवति bhavati
《दा》 dā	repetition	→ ददाति dadāti
《रुध्》 rudh	nasal insertion inside the atom	→ रुणद्धि ruṇaddhi
《सु》 su	nasal extension after the atom	→ सुनोति sunoti

Figure 11.3 — Five activation procedures preserved in completed R̥gvedic verbs. Each procedure changes or extends its atom in a restricted way before the verbal ending completes the form.

Each procedure allows the atom to change only in a specific and predictable way. Atoms that behave alike follow the same preparation. Once a learner understands one procedure, the same knowledge can be applied to the other atoms that share it.

11.3 The R̥gveda Preserves the Wider Verbal Range

The five detailed examples concentrate on the preparation of the atom. Sanskrit can also change the ending attached to a prepared atom.

Direct attachment remains the simplest procedure. The R̥gveda uses अस्ति (*asti*) at RV 1.22.4. The atom 《अस्》 (*as*), meaning *to be*, receives ति without an inserted vowel:

$$《अस्》(as) + ति(ti) \rightarrow अस्ति(asti)$$

The final स् of the atom meets the opening त् of the ending. Nothing enters between them.

Consider «यज्» (*yaj*), meaning *to perform yajña*. The Ṛgveda uses both यजति (*yajati*) at RV 1.26.3 and यजते (*yajate*) at RV 1.31.15. Both verbs retain the same atom and the same inserted अ.

«यज्» (*yaj*) + अ (*a*) + ति (*ti*) → यजति (*yajati*)

«यज्» (*yaj*) + अ (*a*) + ते (*te*) → यजते (*yajate*)

The ति ending belongs to the set called परस्मैपदम् (*parasmaipadam*). The ते ending belongs to the set called आत्मनेपदम् (*ātmanepadam*). A reader does not need the complete ending tables here. These two forms demonstrate the relationship directly. Sanskrit can keep the atom and its preparation unchanged while a different ending gives the completed verb a different grammatical form.

The six singular and plural forms that *laukika* Sanskrit can generate from «इ» show how much information Sanskrit can place inside one completed verb. Sanskrit calls the speaker उत्तमपुरुष (*uttama-puruṣa*), the person addressed मध्यमपुरुष (*madhyama-puruṣa*), and the person or thing being described प्रथमपुरुष (*prathama-puruṣa*). Figure 11.4 places these six forms above five Rigvedic verbs that express present, past, future, command, and possibility or desire.

All six verbs come from the same atom, «इ», *to go*. Each form combines the prepared atom with a compact sonomeric ending. English usually uses a separate pronoun: *I, we, you, he, she, or they*. Sanskrit incorporates person and number into the completed verb. Its preparation and ending together identify whether the action belongs to the speaker, the person addressed, or someone being described. They also distinguish a singular actor from a plural group. A separate pronoun can still identify or emphasize the actor, but the verb does not require one to express person and number.

Encoding person and number inside the verb reduces the sentence's dependence on a fixed word sequence. The verb already identifies the actor and number, so it does not need a fixed position or a separate pronoun merely to carry those facts. Chapter 12 will add the role-endings carried by the other words and show how the complete system permits free word order.

The six forms above demonstrate the engineering through one atom. The Ṛgveda preserves the same person-and-number distinctions across many atoms and also preserves separate forms for exactly two actors. Appendix Part 7 §7.2 records all nine combinations and gives a Vedic passage for every form in its concordance.

The verbs also place actions in different times and modes.

The lower panel establishes that the Vedas preserve the same present, past, future, command, and possibility forms available to the *laukika* domain. Chapter 16 examines the additional forms restricted to the *vaidika* domain.

WHO ACTS?

ONE ATOM: 《 इ 》 (i), TO GO

PERSON	SINGULAR	PLURAL
speaker	एमि <i>emi · I go</i>	इमः <i>imaḥ · we go</i>
addressed	एषि <i>eṣi · you go</i>	इथ <i>itha · you all go</i>
described	एति <i>eti · he, she, or it goes</i>	यन्ति <i>yanti · they go</i>

WHAT THE ACTION EXPRESSES

present भवति <i>bhavati</i> becomes RV 1.17.5	past आसीत् <i>āsīt</i> was RV 10.129.1	future करिष्यति <i>kariṣyati</i> will do RV 1.164.39	command भव <i>bhava</i> become RV 1.1.9	possibility or desire स्याम <i>syāma</i> may we be RV 1.4.6
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Figure 11.4 — The grammatical breadth carried by completed verbs. The upper panel shows how one atom expresses person and number. The lower panel gives Rigvedic examples of five ways in which a speaker can present an action.

The same corpus also preserves participles, infinitives, and gerundives. It preserves questions, negations, commands, and compounds. Chapter 12 follows these forms and the Vedic nominal relations into complete sentences. Appendix Part 7 §7.2 records the wider grammatical inventory and the exact passages behind Figure 11.4.[NOTE: rigvedic-kriya-breadth]

11.4 The Grammar Is Evident in the Vedas

The Vedic passages cited above preserve completed verbs that existed before this chapter separated them for analysis. The passages preserve more than the atoms. They preserve the completed verbs and every sound added or changed during their formation. They also preserve endings that distinguish the actor and number, as well as time and mode. Figure 11.4 demonstrates the same shared architecture through one atom; Appendix Part 7 §7.2 records the complete Vedic concordance.

Repeated forms allow the procedures to be compared. ददाति can be placed beside other verbs that repeat part of their atom. रुणद्धि can be placed beside other verbs

that insert a nasal sound. सुनोति can be compared with other atoms that receive the same kind of extension. A mantra uses the completed verb to express its meaning; it does not pause to teach the formation. Comparison across many mantras reveals that Sanskrit applies the same preparation when it forms the same kind of verb from another atom in that group.

Exact transmission allows the Vedas to serve as a grammatical calibrant. A student in one generation and a student many generations later can examine the same completed forms. Both can check an analysis against एति, भवति, ददाति, रुणद्धि, सुनोति, and thousands of other preserved verbs.

Exact recitation also preserves the pronunciation of those forms. A learner can hear each sonomer, the place in the mouth where it is produced, and the sound changes that occur when the atom and ending meet. The Vedas therefore calibrate both the grammatical construction and its spoken form. Chapter 16 discusses the additional sounds and recitation features restricted to the *vaidika* domain.

The Vedas preserve the sounds, transformations, endings, and grammatical relations that the *laukika* domain can use in new composition. Every new *laukika* word or sentence need not occur in the Vedas. One domain keeps the grammatical examples invariant. The other uses the same language to meet a changing world.

The Vedas must preserve considerable grammatical variety if they are to serve as the calibrant for Sanskrit. A small archive containing one form of one verb could not preserve the language's grammatical breadth. The Rigveda alone fills every person-and-number position and preserves several times, modes, and activation procedures. The remaining Vedas add further forms and styles. Together they allow a learner to examine the language under many different grammatical conditions.

11.5 Pāṇini Made the Analysis Explicit

The Vedas preserve the completed forms. Pāṇini identified the recurring patterns within those forms and stated how a learner can derive them step by step.

His documentation separates three parts that a student might otherwise encounter only as a completed word:

1. the धातुः (*dhātuḥ*), which contains the core meaning;
2. the prepared form of that atom, called the अङ्गम् (*aṅgam*) when it is ready to receive an ending; and
3. the verbal ending, drawn from the set called तिङ् (*tiṅ*).

The preparation between the *dhātuḥ* and the completed *aṅgam* is not identical for every atom. Pāṇini's documentation identifies *dhātavaḥ* that behave alike and records the affix or transformation associated with each group. The grammatical

continuum teaches these recurring groups as the ten गणाः (*gaṇāḥ*) and commonly gathers the intervening operations under the term विकरणानि (*vikaraṇāni*).[NOTE: vikarana-as-column-signature]

The reader has already seen why such a classification helps. «इ» changes its vowel. «भू» changes its vowel and receives अ. «दा» repeats its opening. «रघ» receives a nasal sound inside the atom. «कृ» receives a nasal extension after it. The classification tells a learner which procedure applies to each atom and how the intermediate sounds arise.

Pāṇini did not create एति, भवति, ददाति, रुणद्धि, or सुनोति. Every one of those verbs already occurs in the Vedic corpus. He documented their recurring behavior as precise procedures that a learner could apply in either direction.

Earlier *vaiyākaraṇāḥ* had also analyzed Sanskrit. Pāṇini produced the finest documentation of that accumulated understanding. A learner can begin with a *dhātuḥ*, account for each change, select the required ending, and derive the completed verb. The learner can also reverse those steps, inspect a completed form, and recover the atom and grammatical information within it.

Appendix Part 6 §6.3 examines how far Sanskrit's compact atoms extend across word families and recorded grammatical combinations. Section 6.4 then sets out the ten activation groups and compares those groups with the sonomeric construction of the atoms. This chapter establishes the procedures directly through completed Vedic verbs; the appendix tests their distribution across the larger inventory.

11.6 The Atom Has Become Action

Chapter 10 examined how sonomers form a semantic atom. This chapter has followed that atom into action.

The five detailed verbs demonstrate how Sanskrit prepares different atoms. It may change a vowel, insert a sound, repeat part of the atom, or add material at a restricted position. In every example, the completed verb preserves enough of its construction for a learner to trace it back to the atom.

The wider Vedic examples show how much grammatical information these procedures can express. The endings distinguish who acts and how many act. Other forms place the action in the present, past, or future. Commands and modal forms express further relationships between the speaker and the action. The Vedas preserve these completed forms exactly, which allows them to serve as the calibrant for the language.

A completed verb also repeats Sanskrit's fractal construction. It contains both its ending and its prepared atom, while the atom still contains its sonomers. A learner

can therefore identify the smaller units inside the larger verb.

The *laukika* domain gives this preserved architecture radiant reach. Speakers use it to tell stories and compose poetry. Mathematicians use it to explain proofs, and astronomers use it to record their calculations. Manufacturers and traders can describe their materials, objects, and transactions. Every generation can generate the expressions it needs because the procedures remain available and the calibrant remains unchanged.

The semantic atom has now become a verbal molecule. Chapter 12 follows the next extension: *dhātavaḥ* become several kinds of words, role-endings prepare those words for relation, and the completed forms join as a sentence.